Supplement to: Capability or Optimizer? What Lets an LLM Proposer Leave Its Template Family on Circle Packing

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Section, table and figure references without an S prefix point into the main paper. Each arm here has a row in the paper's Table 4 and a ledger listed in Appendix E; the paper's deviations table (Table 5) covers these arms too. Nothing here is pooled with any row of Table 1.

S1 The pooled zero over six arms

Table 1's zeros pool with three arms outside it that test the same value in regimes the direct rows do not cover — held-out cells, one conditioning step, five loop generations — and the pooled statement is the one an evaluation reader can use:

corpus	regime	clears	where
115 valid program outputs (CC 37, CC2 41, CCS 37)	executed math-only code, two tiers, fresh-draw replication	0	§3.4
63 valid outputs	held-out N, five cells absent from every scoreboard	0	§S9.4
49 valid conditioned outputs	five generations of a registered archive-and-mutate loop	0	§S8.2
63 valid conditioned outputs	one conditioning step, three parent conditions (10^{-9})	0	§S8.1
290 valid outputs (281 at 10^{-9} throughout)	CC, CC2, CCS, CN, MU, L	**0**	—
12 valid program outputs	math-only code, bare pinned API path (not pooled)	0	§S8.3
35 valid program outputs	math-only code, Sonnet tier, pinned path, 120 s (not pooled)	0	§3.5
19 valid program outputs	numpy/scipy code, weak tier, arms CL and CL-W pooled (not pooled here)	0	§3.5
25 valid program outputs	numpy/scipy code, Sonnet tier (not pooled)	**23**	§3.5

Pooled, that is **0 of 290 valid outputs** across six arms, selected by regime and not by outcome: math-only programs at two tiers, cells no scoreboard lists, a conditioning step, an iterated loop. Three things travel with the number. It is not one tolerance: the first three rows count outputs valid at 10^{-6}, the arm MU row counts at 10^{-9}, and arm MU is the one arm where the legs disagree — held at 10^{-6} throughout, the pool reads 315 valid outputs with 18 above the argmax, all 18 from arm MU and every one failing 10^{-9} because the excess is its own overlap (§S8.1). Held at 10^{-9} throughout, which is the tolerance the clearance rule requires, the pool reads 281 valid outputs with 0 above: the three code arms are unchanged (37, 41 and 37 valid at both tolerances, re-scored from the stored programs through the registered scorer), CN 56 of its 63, L 47 of its 49, MU 63. The recount is post hoc bookkeeping over rows already in the study (`recount_1e9.py`); it adds no invocations and moves no verdict. It is not all registered: arms CC/CC2/CCS registered zero-count predictions and arm L's L4 was registered open, but the clearance question was put to arm MU's rows post hoc; arm CH's 14 valid outputs, conditioned on a score table rather than on a parent, are held out of the parent-conditioning row and none of them clears either. And it is not a bound: the three direct-emission escapes of §3.3 and the twenty-three library programs of §3.5 sit outside it by named regime, and the middle-tier escape sits at $N = 31$, a cell five of the six pooled arms also sample. What the zero says is that on this benchmark, at these cells, a weak-tier proposal or a math-only program at either tier does not exceed a value computable in advance, so a discovery claim from those proposers has a number to beat before any search is credited.

S2 The cross-vendor arms in full

§5 states the verdicts; this section carries the chain's verdict table, arm GM3 per cell, and arm V in full.

Vendor	Model	Instrument	Registered test	Verdict
Google	gemma-4-26b	first-party API, 7 $N \times 20$ (41/140 rows routed under amendment; dual accounting)	pooled $\geq$ 30%; cell-level $\geq$ 5; falsifier $\geq$ 4 fails	pooled bar met (57.5%); discriminating cells 11/43 = 26%, **below** the bar; cell-level short (modal in 2 of the 5 scoreable cells, against a bar of 5; falsifier not triggered); all three misses are the *rival construction* (27/43)
Cohere	north-mini-code	free-tier alias, 5 $N \times 5$	V1 $\geq$ 50% of valid on-prediction; V2 rival $\leq$ 10%	**TRANSFERS**, point-estimate (8/12 = 67%; CI includes bar); V2 HOLDS (0/4)
OpenAI (open-weight)	gpt-oss-20b	free-tier alias, 5 $N \times 5$	same	**TRANSFERS**, point-estimate (11/18 = 61%; CI includes bar); V2 HOLDS (0/9)
Google	gemma-4-31b	free-tier alias, 5 $N \times 5$	same	**DOES-NOT-TRANSFER** (0/15); V2 HOLDS (0/3)
10 further aliases	(NVIDIA, Liquid, Poolside, inclusionAI, others; one unattributed)	free-tier	floor: $\geq$ 10 of 25 invocations valid	BELOW-FLOOR; failure taxonomies logged, no claim in either direction

S2.1 Arm GM3 per cell

Arm GM3. Arm GM3 asks whether the anchoring law is a fact about the original lineage or about weak-tier models as a class, by re-running the unconditioned-call protocol against a second vendor's open-weight family (gemma-4-26b-a4b-it) through its first-party API: seven N cells, twenty samples each, registered predictions and falsifier fixed before sampling (prereg chain in Appendix E; the instrument's history — a first attempt whose 4,096-token budget was consumed entirely by the model's deliberation channel, resolved by a registered re-run at 16,384 — is in §7, and is itself a finding about output-budget confounds in cross-vendor comparisons). Per cell, recomputable from `arm_gm_gm3_checkpoint.jsonl` by `arm_gm3_analysis.py` with no arguments:

N	predicted (4 dp)	Gemma modal sum	mode freq	valid	on-prediction	MODE-MATCH
13	1.6250	1.7761	7/9	9	0	no (misses upward)
17	2.0518	**2.0518**	19/20	20	20	**yes**
21	2.1000	2.2589	9/18	18	8	no (misses upward)
31	2.5833	2.7485	11/16	16	3	no (misses upward)
35	2.9167	**2.9167**	15/15	15	15	**yes**
37	3.0345	—	—	2	0	UNSCOREABLE
43	3.0714	—	—	0	0	UNSCOREABLE

Three registered outcomes (Figure S1). **The pooled prediction held**: 46 of 80 valid packings (57.5%) land within 2×10^{-3} of the value predicted by §2's rule, against a registered 30% bar — for a rule fitted entirely on another vendor's models, before this family was ever sampled. The registered metric pools all cells, and the pass rides on that: by §S9.2's own convention — arm F's pooled rate excludes non-discriminating cells *because prediction equals the family argmax there* — the decomposition must be stated. 35 of the 46 on-prediction packings sit in the two non-discriminating cells (20/20 at $N = 17$, 15/15 at $N = 35$); on the three scoreable discriminating cells the rate is 11 of 43 (25.6%), below the bar the pooled metric passes. The registered verdict stands as registered; what it measures at the discriminating cells is the next finding. **The cell-level prediction did not hold**: the predicted value is modal in 2 of 5 scoreable cells, short of the registered 5, though the registered falsifier (failure in $\geq$4) also did not trigger — and where the family locks on, it locks on

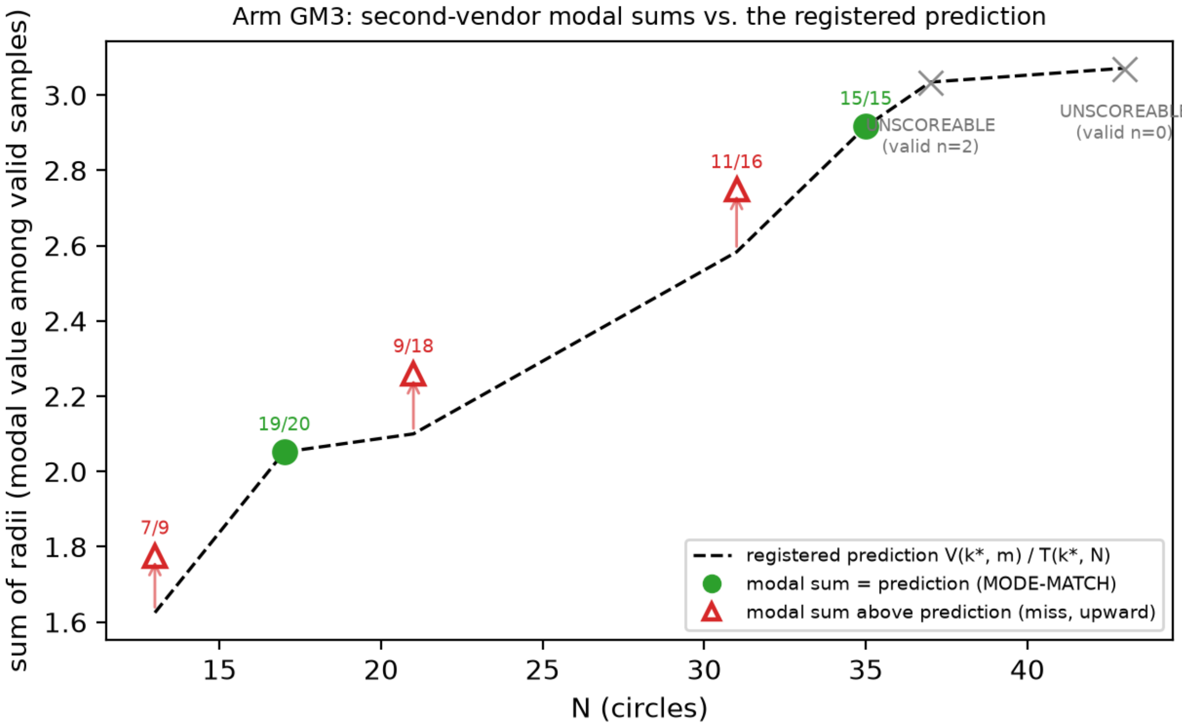

Figure S1: *Arm GM3 modal sums against the registered prediction.* Filled green circles match the predicted mode (the dashed line is $V(k^*, m)$ at extend cells and $T(k^*, N)$ at truncate cells) (frequencies annotated); open red triangles miss — every miss lands above the prediction, on the rival argmax (the modal packings carry the rival's two-radius structure, 27/27 — §5); the two largest cells are UNSCOREABLE under the registered <3-valid rule. Regenerates via `fig_gm3_anchoring.py` from `arm_gm3_report.json` (itself replayed from the raw ledger by `arm_gm3_analysis.py`, no arguments).

almost completely. That pair of bars leaves a dead zone: cell-level outcomes of 2, 3 or 4 confirm nothing and falsify nothing, and the observed outcome landed in it. The registration is at fault, not the result; we report the verdict it defines and note that a registration with a gap this wide buys less than it appears to. **The structural signature came in under its bar**: 20 of 46 on-prediction samples (43%, bar: half) carry the exact two-radius grid-plus- filler decomposition. The instrument is blind by construction at truncate cells, whose prediction is single-radius: all 20 signatures come from $N = 17$ (20 of 20), the one extend cell among the matches, while $N = 35$'s 15 on-prediction truncations carry no filler to detect (0 of 15). The registered verdict stands as registered; read per eligible cell, the construction matches wherever it can be seen.

The three cell-level misses are not noise, and "misses upward" understates them: **the family's modal output at every discriminating cell is the rival argmax itself** — 1.7761, 2.2589 and 2.7485 against rivals 1.7761424, 2.2588835 and 2.7485281 — and a post-hoc structural diagnostic (disclosed as such; `diagnostics_gm3_rival.py`, read-only over the ledger) finds all 27 of 27 rival-valued packings carrying the rival's exact two-radius decomposition, the (k^*-1) grid radius with the (k^*-1) filler radius. The registered signature instrument could not see this: it tests radii against the *predicted* order k^*, so it is structurally blind to (k^*-1) constructions — which is why the 43% signature figure above coexists with a 100% rival-structure rate on the miss side. In §2.3's terms, gemma-4-26b performs precisely the drop-to-(k^*-1)-and-fill move that "a proposer with any local search would" perform and that the original family performs in 2 of 57 valid samples: 27 of 43 valid discriminating-cell packings (63%) are the rival construction. Read against §7's tractability alternative, the anchor value marks where construction-by-mental-arithmetic lands, and a family with more arithmetic reach executes the harder in-family construction rather than truncating — same recipe family, other branch. Validity collapsing at the two largest cells (2 then 0 valid of 20) bounds the family's competence window on this task; both cells are UNSCOREABLE under the registered rule (< 3 valid samples, fixed in the GM-chain registration before sampling) and claim nothing.

S2.2 Arm V in full: cross-vendor zero-shot under free-tier serving

§5 is the contract; this is the report.

Arm V. Arm GM3 varied the vendor while holding the instrument close to laboratory grade — first-party API, pinned temperature, explicit output budget, seven cells at twenty samples. Arm V varies vendor and serving class at once, under the same byte-identical arm-F prompts both arms inherit (SHA-256 per N restated in the registration and recomputed at call time, aborting on mismatch; the GM-chain registration pins the identical hashes), at five cells and five samples per (model, N), against free-tier aliases across seven attributed vendors (`arm_v_preregistration.md`, committed before sampling; three amendments, each committed before the sampling it governs, expanded the original three-alias design to thirteen aliases across two serving paths — the opencode CLI and OpenRouter chat-completions — and raised `max_tokens` to 24,576 after hidden-reasoning truncation emptied three models' replies). Serving disclosures per the companion paper's protocol: aliases on managed paths, `sampling_params: null` on every row, per-row upstream-provider strings logged where the path reports them, every invocation appended live to `arm_v_candidates_raw.jsonl` — 877 invocations, failures and refusals included. Scoring is `arm_f_repro.py`'s parse/validate/classify unchanged, tolerances registered in arm F before its sampling.

Registered floor: ≥10 of a model's 25 registered invocations valid at 10^{-6}; below it, no anchoring claim in either direction. Execution retried each invocation slot until content arrived, per amendment 3's registered transport-repair and accounting rule, so the ledger holds many more attempt rows than slots — gemma-4-31b's 25 slots, for instance, took 190 attempt rows, 165 of them transport failures — and the floor is evaluated on the 25 slots, not the attempts. Ten of thirteen aliases finished BELOW-FLOOR — including all three originally-registered opencode aliases, whose logged failure taxonomies are transport-dominated for one (CLI build banners, timeouts, "Unknown model" errors as the free tier rotated its catalogue) and roughly half transport, half parse-failure for the other two — plus one alias returning all 108 of its attempts without content. Three aliases cleared the floor, and their registered verdicts split:

alias (vendor)	valid	V1 on-prediction	V1 verdict (bar: ≥50%)	V2 rival-argmax (bar: ≤10%)
`north-mini-code` (Cohere)	12	8/12 [39–86%]	**TRANSFERS** (point est.)	0/4 → **HOLDS**
`gpt-oss-20b` (OpenAI, open-weight)	18	11/18 [39–80%]	**TRANSFERS** (point est.)	0/9 → **HOLDS**
`gemma-4-31b-it` (Google)	15	0/15 = 0%	**DOES-NOT-TRANSFER**	0/3 → **HOLDS**

This is the registration's most informative branch pair. Anchoring under this instrument is not an artifact of the original family: two further vendors' models land the registered construction values at 61–67% of valid outputs, point estimates above the 50% bar.

Both passes are weaker than those percentages suggest, in three ways we state rather than leave to the reader. The bracketed Wilson 95% intervals include the bar, so each verdict is a point-estimate pass at $n = 12$ and $n = 18$. Three of thirteen aliases were evaluable at all; the other ten finished BELOW-FLOOR and claim nothing in either direction. And the pooled rate spans cells that cannot discriminate anchoring from family search. Decomposing onto the discriminating cells alone — the split this section demands of the GM chain, and symmetry demands here — gives 3 of 4 and 5 of 9. Unlike GM3 the decomposition does not reverse either verdict, but it shows how little of each pass sits where prediction and family argmax differ.

§S9.2's template-shape null belongs here too, and applying it costs the arm its strongest reading. Against that null a proposer lands on-prediction one time in three. The pooled rates clear it — exact upper tails 0.019 and 0.014 — but they clear it on cells where prediction equals the family argmax. At the cells that can discriminate, neither pass separates: 3/4 gives 0.111 and 5/9 gives 0.145, and GM3's 11/43 gives 0.895, below the null. So the honest cross-vendor statement is that two further vendors are *consistent with* the behaviour and that at $n = 4$ and $n = 9$ the discriminating evidence does not separate them from a shape-uniform proposer. Recomputed in `cross_vendor_null.py`; post hoc, like the null itself.

The strong form holds everywhere evaluable *within this arm*: zero rival-argmax emissions in 16 discriminating-cell validities. The registration (`arm_v_preregistration.md`) recorded the original family's prior for the same statistic as 0/21 over $N \in \{13, 31\}$ only, a narrower scope than §S9.3's 2/23 whose two rival hits both sit at $N = 21$, so the two corpora agree at 0 on the registration's own cells. The standing exception is gemma-4-26b under the GM-chain instrument, whose discriminating-cell mode *is* the rival.

The transfer's point estimates sit at or below the in-family rates: 61–67% on-prediction against 56–76% in-family at the same five cells (§S9.2). The ranges overlap, so reduced concentration is a directional reading these n cannot separate. Directionally lower concentration at different vendors, on byte-identical prompts, landing on the same modal value is what a shared attractor predicts; equal concentration would be harder to separate from shared training text. And it is not universal: gemma-4-31b, clearing the same floor, lands the prediction never. Its valid packings are unanchored and low (bests 0.88–1.85 against predictions 1.63–3.03) — not the rival, not the template, just weaker de-novo construction. Under the registration's falsification map, the family-specific branch fires — any model clearing the floor with DOES-NOT-TRANSFER makes the closed form a family-bounded property, stated in §3.3 with data rather than a caveat — while the map's all-transfer branch fails on gemma alone, so the reading that anchoring is an artifact of the original family is defeated by the same table that bounds its generality.

The gemma pair needs stating precisely, because this family now appears on both sides, and the prompts are identical across the two arms — what differs is parameter count, serving path and sample budget, not the elicitation. Under the GM-chain instrument (first-party, pinned temperature, 7×20), gemma-4-**26b** met the pooled bar at 57.5% while its discriminating-cell rate (26%) fell below it (the mixed-serving accounting; first-party-only, 59.0% — §7's registered deviation applies to this figure) while emitting the *rival* construction at 63% of discriminating-cell validities; under arm V's free-tier instrument (5×5), gemma-4-**31b** through a rate-starved alias (Google AI Studio upstream; 27–38 of each cell's attempts returned as shared-pool 429s before content arrived) lands neither the prediction nor the rival — its valid packings are weaker de-novo constructions. The pair bounds the family's behaviour rather than contradicting itself — one member sits *above* the template's execution reach (far enough to build the drop-and-fill rival the original family cannot), the other lands below it — but model variant and instrument quality change together across the two measurements, so which difference carries the split is not attributable from this pair; either way, no single-model measurement would have seen the boundary.

S3 The `opus_alias` arm: the dual accounting

Appendix D carries the ladder table and the failure taxonomy; the two post-hoc decisions and the two corpora behind the ladder's first rung are here.

Two decisions here were made after seeing the data and are recorded as deviations (Table 5). P-O1, P-O2 and P-O4 are reported **not evaluable**, on the grounds that a validity collapse of this size makes an on-prediction tier comparison uninterpretable; P-O3 is trivially satisfied on 4/4 valid samples; the registered disconfirmation — regression toward the trap — did not occur. De-registering three of four predictions in the one arm that disconfirmed is exactly the move preregistration exists to constrain, which is why it is tabled rather than narrated. And the arm was initially excluded from the ladder, then included once its results were known; we follow the later decision and report the ladder both ways (Table 6) — with the arm 83% → 100% → 13% on matched cells, without it 83% → 100% plus the qualitative shift from truncation to perturbed multi-radius hybrids, which is already the publishable contrast.

Two rungs, two corpora, and they are easy to confuse. The 83% is the 60-row matched-cell ledger at $N \in \{13, 21, 31\}$. The 78% that circulates beside it is the bare arm's own 45-row ledger over all seven weak-tier cells — a different corpus and a different cell set, not a fourth rung. The 10^{-9} ladder reads 64% → 90% → 13%, where the 64% is again the 45-row seven-cell ledger and its matched-cell counterpart was not computed, so the two ladders are not directly comparable at their first rung either. Both rows appear separately in Table 6 and neither figure is quoted without its corpus.

S4 Arm M in full: the filler branch fails at $m > 1$; the fifth k confirms

The registered verdicts, Table M and the full report of the extend-cell arm are here. Two gaps remained after the square arm: (i) $V(k,m)$'s filler branch was empirically supported only at $m \in \{0,1\}$ — the converge cells $N = 17$ and 37 both have $m = 1$ — and (ii) the $k = 8$ trap zone appeared in Figure S3 but was never sampled. Arm M closes both gaps and was registered before sampling: $n = 15$ bare weak-tier invocations at each of $N = 20$ ($k^* = 4$, $m = 4$, predicted $V(4,4) = 2.2071068$), $N = 30$ ($k^* = 5$, $m = 5$, $V(5,5) = 2.7071068$), $N = 41$ ($k^* = 6$, $m = 5$, $V(6,5) = 3.1725890$) and $N = 57$ ($k^* = 8$, trap, $T(8,57) = 3.5625000$, rival $V(7,8) = 3.7366935$), with an explicit tie-stated falsifier for each half.

The falsifier for the filler branch triggered at 3 of 3 cells. The registered predictions P-M1–P-M3 are disconfirmed: not one of the three converge cells has $V(k^*,m)$ as its modal output. The model does not extend a k^*-grid with fillers; it moves *up* to the (k^*+1) grid and truncates or exactly fills it — modal outputs $2.0 = T(5,20)$ (12 of 15 valid; $N = 20$ is the 5×4 rectangle exactly), $2.5 = T(6,30)$ (11 of 14; 6×5 exactly) and $2.9285714 = T(7,41)$ (6 of 7; 7×6 minus one). Exactly one sample in 36 valid converge-cell rows emitted the registered construction — a textbook $V(4,4)$, all four fillers at $(\sqrt{2}-1)/8$. Per the registered falsifier wording, $V(k,m)$ **is restricted to the $m \leq 1$ support previously observed**, and the branch rule's extend arm is disconfirmed at $m \geq 4$; the intermediate "places some fillers" outcome named in the registration did not occur (filler count 0 in 35 of 36 valid converge rows). The emergent regularity is arithmetic, not geometric: where N factors as $a \times b$ with a, b near $\sqrt{N}$ the model emits that exact rectangle of uniform circles ($20 = 5 \times 4$, $30 = 6 \times 5$), which is further evidence for §7's arithmetic-tractability alternative — rectangle factorization is mentally computable where filler tangency is not.

P-M4 confirmed. At $N = 57$ the modal valid output is $T(8,57) = 3.5625000$ (6 of 10 valid at 10^{-6}), the $k = 8$ grid truncated to 57; 0 of 10 valid samples reached the rival $V(7,8)$, against a registered rival prediction of 0. The branch rule's truncate arm now has five confirmed k (4, 5, 6, 7, 8) and Figure S3's $k = 8$ zone is sampled. Validity by cell: 15/15, 14/15 (one parse failure — fraction literals), 7/12 and 10/15 at 10^{-6}. Three $N = 41$ invocations died in the runtime before reaching a model and are excluded under the rejection rule registered for this arm in registration — registered precisely because the analogous arm F exclusion was not (§8) — and counted here: the cell is $n = 12$ sampled of 15 launched. Raw rows verbatim in `arm_m_collect.jsonl`; scoring in `arm_m_analysis.py`, conventions identical to arm F.

Table M: Arm M, forecast versus outcome (all four registered cells).

N	k^*	Branch	Registered prediction	Modal valid output	Freq	Valid / sampled	Verdict
20	4	extend, $m = 4$	$V(4,4) = 2.2071068$	$T(5,20) = 2.0000000$ (5×4 exact)	12/15	15/15	P-M1 **disconfirmed**
30	5	extend, $m = 5$	$V(5,5) = 2.7071068$	$T(6,30) = 2.5000000$ (6×5 exact)	11/14	14/15	P-M2 **disconfirmed**
41	6	extend, $m = 5$	$V(6,5) = 3.1725890$	$T(7,41) = 2.9285714$ ($7 \times 6 - 1$)	6/7	7/12	P-M3 **disconfirmed**
57	8	truncate	$T(8,57) = 3.5625000$, rival count 0	$T(8,57) = 3.5625000$	6/10	10/15	P-M4 confirmed; rival 0/10

Exactly one $V(k^*,m)$ construction appears in 36 valid converge-cell rows; filler count is 0 in 35 of 36. Three converge-cell rows exceed the recipe family's best *upward*, of which two clear it under §2.4's rule — two at $N = 20$ (staggered rows of four at $r = 1/9$, sum 2.2222222, against the family best $V(4,4) = 2.2071068$; both valid at 10^{-9}, all 15 of that cell's valid rows are) and one at $N = 30$ ($r = 1/11$, sum 2.7272727, against $V(5,5) = 2.7071068$; valid at 10^{-6}, not at 10^{-9}, so it does not clear) — hexagonal layouts the family does not contain, found by a post-hoc sweep (not preregistered) run after a fourth-round check; they are not the modal output at either cell and do not bear on the discriminating-cell claims. Regenerates from `arm_m_collect.jsonl` via `arm_m_analysis.py`.

The net effect on the paper's claim is a sharpening. Arm M does not rescue $V(k,m)$; the closed form now reads "$k^* = \mathrm{round}(\sqrt{N})$ names the grid family; at $k^{*2} > N$ the model truncates that grid (five k confirmed); at $k^{*2} \leq N$ it extends only to $m \leq 1$ — beyond that it prefers the $(k^*{+}1)$ rectangle, and $V(k,m)$ does not describe it."

S5 Rectangle transfer (arm G)

S5.1 Transfer of the rule

For a $1 \times a$ rectangle the template has two parameters: $q^* = \mathrm{round}(\sqrt{N/a})$ columns across the width, $p^* = \mathrm{round}(\sqrt{N \cdot a})$ rows across the height. At $a = 1$ both collapse to $\mathrm{round}(\sqrt{N})$ — the same rule with the aspect ratio put back, not a new rule fitted to new data, and no rectangle model output existed when it was written. It was verified against an independent LP at 213 configurations, drift below 10^{-9}, including a cross-domain check: at $a = 1$ with $p = q = k$ the square file's closed form must equal this file's LP. Shape mismatch grows as a leaves 1 — over $N = 10 \ldots 45$ the count of N whose predicted shape differs from the optimal one rises from 12 at $a = 1.0$ to 23 at $a = 1.5$, 20 at $a = 2.0$ and 23 at $a = 3.0$.

We probed the two sharpest cells with sixteen invocations (eight per cell) in a container none had been given before: $N = 19$ at $a = 3$ (predicted 3.1666667, rival 3.5749194) and $N = 25$ at $a = 2$ (predicted 3.1250000, rival 3.4832492). With all sixteen scored, **5 of 11 valid proposals landed on the predicted value and 0 of 11 reached the rival**.

We characterize all eleven here. On-prediction: two at $N = 19/a = 3$ (3.1666666667 and 3.16666673, the latter agreeing to six decimals not seven) and three at $N = 25/a = 2$ (3.125 exactly). Off-prediction at $N = 25/a = 2$: two samples emitted a uniform 5×5 grid at $r = 0.1$ summing to 2.5000000 — the *square* rule with no aspect correction, $k = \mathrm{round}(\sqrt{25}) = 5$ — plus one at 3.0 and one at 3.151875, the latter above the 3.125 prediction from outside the family and below the rival. At $N = 19/a = 3$, **two** further samples exceeded the prediction from outside the family, 3.45 and 3.5, both below the rival.

We therefore report the rectangle as **partial out-of-sample support, not confirmation**: $5/11 = 45\%$, Wilson 95% CI $[21\%, 72\%]$, and a proposer choosing uniformly among the three or so plausible template shapes would land on-prediction about a third of the time, so 5/11 does not separate cleanly from that null. Two qualifications: validity degrades in the tall container (4/8 at $a = 3$ against 7/8 at $a = 2$, three of four $a = 3$ failures being overlaps), a separate finding and a confound on that cell at $n = 8$; and the rectangle ledger carries no prompt hashes, no run dates and no proposer fields, so the arm with the strongest transfer claim has the least provenance (§8).

S5.2 Negative result: the closed form does not survive the move to rectangles

The rectangle generalizes the *rule* but not the *formula*, and we keep the failure because it is what a verification gate is for. In the square every interior vertex has four identical neighbors, so one expression covers every filler. In a $1 \times a$ rectangle, with half-spacings $h_x = 1/(2q)$ and $h_y = a/(2p)$, a filler is also capped by horizontal and vertical spacing to *adjacent fillers*. Those constraints are inactive when $h_x = h_y$ — why the square case could not have revealed them — and bind only when neighboring vertices are occupied, so the cap depends on m and on which vertices a construction uses, and no expression in (p, q, m, a) reproduces it. The LP gate caught this on the first run: at $a = 1$, $p = 2$, $q = 4$, $m = 1$ the natural generalization $\texttt{rf} = \min(\texttt{diag}, h_x, h_y)$ returns **1.125** against a true **1.1545085**, capping against a neighboring filler that does not exist. We retain closed forms only where provably exact (full grid and truncated grid) and use the LP as the value oracle for the extend branch, so what §S5.1 tests is the LP-backed prediction.

S6 Elicitation and faithfulness (arm T)

This arm is secondary to the paper's question and its result leans negative: the registered validity prediction fails, the concentration effect survives only as a fragile directional finding, and the faithfulness audit checks description against artifact, not against process.

Table S1: Forecast versus outcome, both containers.

Cell	Branch	Predicted	Rival argmax	Valid / sampled	On-pred / valid	Rival
N=13 (sq)	truncate	1.6250000	1.7761424	4/5 · **18/20**	3/4 · **10/18**	0
N=17 (sq)	extend	2.0517767	†	4/5	3/4	†
N=21 (sq)	truncate	2.1000000	2.2588835	7/10 · **15/20**	4/7 · **12/15**	2
N=31 (sq)	truncate	2.5833333	2.7485281	5/5 · **17/20**	5/5 · **13/17**	0
N=35 (sq)	truncate	2.9166667	†	4/5	3/4	†
N=37 (sq)	extend	3.0345178	†	4/5	3/4	†
N=43 (sq)	truncate	3.0714286	3.2416246	7/10	6/7	0
N=19, a=3	truncate	3.1666667	3.5749194	4/8	2/4	0
N=25, a=2	truncate	3.1250000	3.4832492	7/8	3/7	0

Plain entries are the original 45-invocation bare arm (§S9.3's corpus); **bold** the same cells over the full 85-row bare ledger after §S6 added 40 rows. † marks non-discriminating cells (prediction equals family argmax), excluded from rival denominators. Square discriminating cells: 18/23 and 2/23 on the original corpus; 41/57 and 2/57 on the full ledger. Rectangle: 5/11 and 0/11. Sources: `arm_f_repro.py` over `arm_f_candidates_v2.jsonl`, `arm_g_rect.py` over `arm_g_candidates.jsonl`.

S6.1 Design and bundling disclosure

A ten-versus-ten pilot at $N = 21$ compared the bare prompt against a variant asking the proposer to name its construction on a leading `METHOD:` line: validity rose from 7/10 to 10/10 and the higher-scoring rival disappeared, from 2 of 7 valid bare samples to 0 of 10 trace samples. Before any scaled sample was drawn we registered four predictions and an explicit falsifier in `arm_t_preregistration.txt` (sha256 `ab7900a8...`), disclosing that the pilot's trace prompt had drifted from the bare template beyond the method line — it omitted the `[0,1]x[0,1]` tokens and reworded the output-format line — so the pilot's intervention was method-line-plus-rewording, bundled. Pilot samples are never pooled with `trace_v2`.

The scaled prompt, `trace_v2`, is itself a **bundled prompt-format-and-trace-request** intervention, though a near-minimal diff: one inserted `METHOD:` line, plus `"After the METHOD line, "` prepended to the output line and `"no other text"` changed to `"no other text after the list"`. That second change is not cosmetic — §S6.2 shows it doing measurable work — so the measured effect is *method-line-plus-output-line-rewording*, not the method line alone: a weaker version of the confound for which the pilot was discarded, held to the same standard.

The scaled arm ran 100 new invocations, bringing both arms to 20 per cell at $N \in \{13, 21, 31\}$. The study corpus accumulates as tabulated below. Each arm is scored under its own registration; the running total is the figure quoted elsewhere in the paper, and with arms P, CL, CCP, CL-W and P-D the study corpus now stands at 1500 invocations.

arm or group	section	invoca
square (bare 85, trace 70, sonnet 30, `opus_alias` 30)	§3.3, §S6	
rectangle	§S5	
M	§S4	57 sampled of 60 laun
MU and CH	§S8.1, §4.1	180, non
CC, CC2 and CCS	§3.4	135, non
CN, held-out N	§S9.4	
CP, RP and PP	§S9.5	270, three rejections cou
L, iterated loop	§S8.2	120, two rejections cou
P, pinned path	§S8.3	225, six refused by the serving
CL, library code	§3.5	90, four refused by the serving
CCP, isolating arm	§3.5	45, one in-flight refusal re
CL-W, weak-tier top-up	§3.5	45, non
P-D, runtime component	§S8.3	27 of 30 collected, closed there by amendment, three refused by c

Table S2: Paired trace grid (scaled arm only).

N	Arm	n	Valid (parse / geom. fail)	On-pred / valid	Rival
13	bare	20	18 (0 / 2)	10 (56%)	0
13	trace_v2	20	18 (0 / 2)	16 (89%)	0
21	bare	20	15 (1 / 4)	12 (80%)	2
21	trace_v2	20	18 (0 / 2)	16 (89%)	0
31	bare	20	17 (2 / 1)	13 (76%)	0
31	trace_v2	20	17 (0 / 3)	14 (82%)	1
pooled	bare	60	50 (3 / 7)	35 (70%)	2
pooled	trace_v2	60	53 (0 / 7)	46 (87%)	1

The square row is 85 bare (45 original + 40 new), 70 trace (10 pilot + 60 trace_v2), 30 sonnet and 30 `opus_alias`. The CP/RP/PP row is 75 offset-container, 90 recall, a 15-invocation registered positive control and 90 paraphrase invocations. The cross-vendor chain sits outside this ladder because it is scored under its own registrations: 420 GM-chain invocations (§5, §7) and 877 arm-V attempt rows over the 325 registered invocation slots, 314 of which carry rows (§5). One disclosure: **20 of the 60 bare samples are pre-existing arm-F rows** ($N = 13$ ids 1–5, $N = 21$ ids 1–10, $N = 31$ ids 1–5) whose results were known when P-T1–P-T3 were written; only trace_v2 is fully fresh (§S6.4). Scoring used `arm_f_repro.py` unchanged with the registered 2×10^{-3} window; the Fisher exact test comes from the hypergeometric tail in `arm_t_analysis.py`, checked against a reference.

S6.2 Registered outcomes

Preregistered criteria are evaluated exactly as registered, and unregistered alpha thresholds are applied nowhere.

P-T1, validity: NOT confirmed. P-T1 is the one prediction that registered an alpha ("validity $\geq$ bare at EACH N, and pooled one-sided Fisher $p < 0.05$"). Direction held at all three cells, but the pooled test gives $p = 0.30$: 53/60 (88%) trace_v2 against 50/60 (83%) bare, a difference whose detection needs several hundred samples per arm at 80% power — a failure to detect, not a demonstration of no effect. The direction that did hold is not geometric: bare had **3 parse failures and 7 geometric failures**, trace_v2 **0 parse and 7 geometric**, so among parsed samples geometric validity is 50/57 (87.7%) versus 53/60 (88.3%). The P-T1 direction is format compliance, exactly what the reworded output-format line manipulates; validity is reported split by cause from here on.

P-T2, rival suppression: CONFIRMED as registered (directional, no alpha): 1 of 53 valid trace_v2 against 2 of 50 valid bare. Counts this near zero carry no inferential weight and we claim none (Fisher $p = 0.48$, carrying no decision). One trace_v2 sample at $N = 31$ hit the rival 2.7485281 to within 3.6×10^{-8} — the first weak-tier rival hit at $N = 31$ in any arm, and formerly one of the scorer's mismatches (§S6.5).

P-T3, anchor concentration: met as registered (directional), inferentially fragile. Among valid samples, 46 of 53 trace_v2 (87%) landed on the registered prediction against 35 of 50 (70%) bare. The p-value it never carried is 0.0325, one-sided uncorrected (§S6.4).

P-T4, faithfulness: confirmed under a corrected scorer. The registered scorer gives 38 of 41 scoreable claims matching (93%) against a 90% threshold; blind hand-adjudication under a rubric frozen before rescoring gives 54 of 56 (96.4%). §S6.5 reports both.

Per-cell one-sided Fisher on on-prediction (pooled bars in Figure S2): $N = 13$ $p = 0.030$; $N = 21$ $p = 0.41$; $N = 31$ $p = 0.50$. Pooled $p = 0.0325$; Holm threshold over the registered family ($m = 3$ tested with p-values) is 0.0167. The pilot (10 invocations at $N = 21$, 10/10 valid, 9/10 on-prediction) is reported separately and never summed into any total. Regenerated by `arm_t_analysis.py` over `arm_f_candidates_v2.jsonl`.

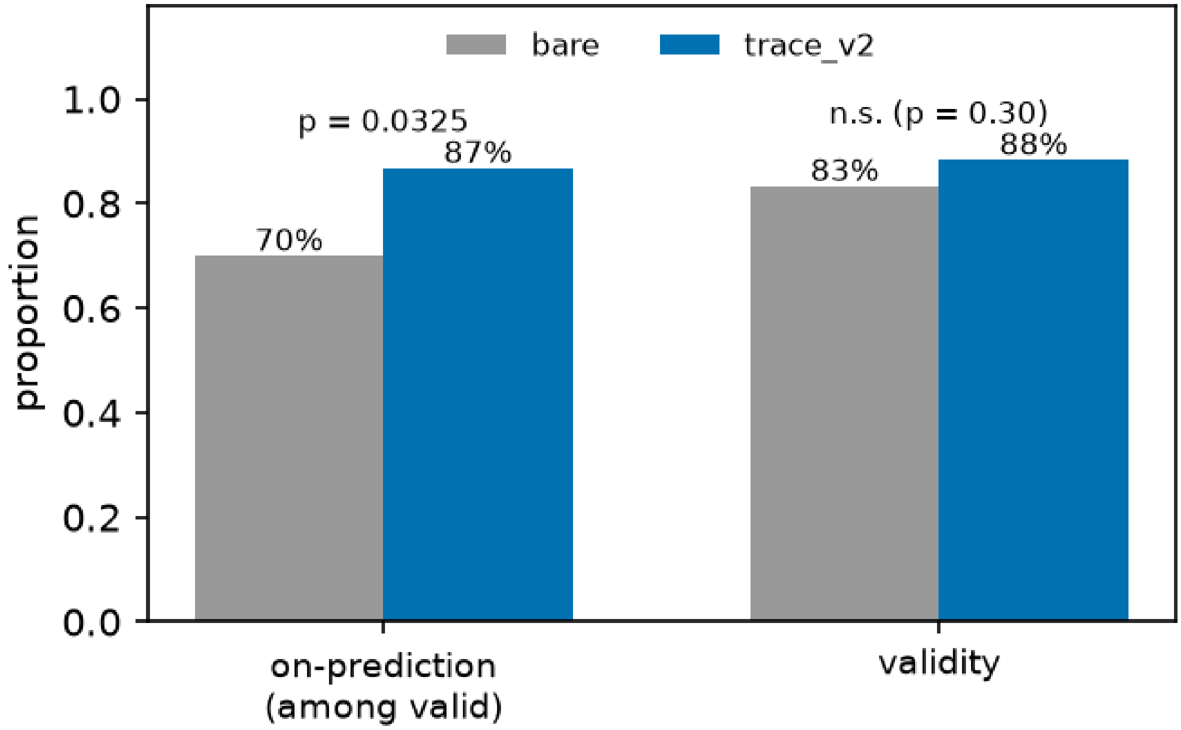

Figure S2: *Bare versus `trace_v2`, pooled over cells.* The annotated $p = 0.0325$ is uncorrected and fails the Holm threshold (§S6.4); per-cell values are in the text. Caption caveat: the bars pool the bare arm's two collection waves, which §S6.4 shows drift materially, so the pooled bare bar is a mixture; the same-wave comparison in §S6.4 is the wave-controlled picture.

S6.3 The registered falsifier, both readings

The registered falsifier reads: *"if trace_v2 validity `<=` bare at 2+ of 3 N, the pilot effect was the bundled rewording, not the method-line request — reported as such, not reframed."* Observed validity: $N = 13$, 18/20 both arms — **tie**; $N = 21$, trace_v2 18/20 against bare 15/20 — trace higher; $N = 31$, 17/20 both arms — **tie**. A tie satisfies `<=`, so under the registered wording the falsifier is **triggered at 2 of 3** N, and we report it as triggered; the code instead evaluated direction as `t_rate >= b_rate`, strict `<`, under which it fires at 0 of 3 cells. That convention appears nowhere in the preregistration, was chosen at analysis time, and read the same tie favorably twice — as "direction held" for P-T1 and as "not a falsifier cell" here.

We name the failure mode **operationalization drift between preregistration text and analysis code**: registered prose leaves boundary behavior — ties, inclusive versus exclusive bounds, rounding — implicit, and the scorer must choose after the data exist; the preregistration literature (§6) discusses what is locked, not this tie-handling layer. Remedies, free before sampling and impossible afterward: register the comparison as executable code, or state the tie convention in the registration text. `arm_t_analysis.py` now prints both readings and names the registered tie-inclusive one authoritative. Under that reading **the pilot's validity effect is attributed to the bundled rewording, not to the method-line request**, as the falsifier clause instructed; the §S6.2 parse-versus-geometric split points the same way, and P-T1 fails under both readings regardless.

S6.4 Limits of P-T3: what the concentration result will and will not carry

P-T3 met its registered directional criterion. It is not a flagship result.

No registered alpha. P-T3 registered direction only; the p-value is a post-hoc addition.

Multiplicity. Of four registered predictions, three are reported with p-values. Holm sets the first threshold at $0.05/3 = 0.0167$; $p = 0.0325$ fails it, as it fails Bonferroni at any $m \geq 2$ and Benjamini–Hochberg FDR at $q = 0.05$. Two-sided Fisher is $p = 0.0537$. P-T3 is nominally significant uncorrected and is not under family-wise error control.

Concentration in one cell. Per-cell tests give $p = 0.030$, 0.41, 0.50 (Table S2); the pooled 17-point gap is a 33-point gap at $N = 13$ averaged with 9- and 6-point gaps elsewhere, driven by a *low bare rate at* $N = 13$ ($10/18 = 56\%$) rather than a high trace rate — the trace rate there (89%) equals that at $N = 21$. Dropping $N = 13$, the comparison is 30/35 vs 25/32, $p = 0.31$.

Wave confound. The bare arm mixes two collection waves; trace_v2 is one. Splitting bare on the registered id boundary, with no intervention applied to either half, on-prediction is: $N = 13$, 3/4 old wave vs 7/14 new; $N = 21$, 4/7 vs 8/8; $N = 31$, 5/5 vs 8/12 — the control arm's own between-wave drift is comparable to the effect attributed to the manipulation. (An earlier draft printed 2/4 vs 10/11 at $N = 21$ and a 25/37 pooled figure — arithmetically impossible against §S6.1's id boundary; corrected here from the ledger.) Restricting to one wave, new-wave bare is 23/34 (68%) against trace_v2's 46/53 (87%), but **this stratified comparison is post hoc, not preregistered, no confirmatory weight** — a diagnosis, not a rescue: wave is a confound *in the registered analysis*, which cannot separate it from the manipulation. The design fix, interleaved collection in one session, was not run.

What we therefore claim. A minimal method-naming request does not make the proposer better at geometry and does not measurably change which rare constructions it reaches; in the registered direction it concentrates output onto the anchor, but that concentration fails correction, is carried by one cell, is confounded with collection wave, and is bundled with an output-format rewording that demonstrably affects parse compliance. Studies collecting process descriptors *by asking for them*, including descriptor-driven quality-diversity pipelines where the descriptor is a requested self-report, should treat this as a reason to check, not a coefficient to apply. The pilot also showed a *third* effect in the P-T3 direction at the same cell (trace 9/10 on-prediction against bare $N = 21$ at 4/7, one-sided $p = 0.16$), so P-T3 replicated a pilot signal rather than testing a blind prediction — a different status, and the prereg's pilot disclosure covered the prompt drift only.

S6.5 Faithfulness with ground truth

Claim and artifact sit in the same completion and the artifact is fully determined, so a method line is checkable against the emitted coordinates. The original automated scorer, `arm_f_repro.trace_faithfulness()`, returned 38 of 41 scoreable claims matching (93%) against the registered 90% threshold, but nine of its twelve exclusions were regex coverage gaps rather than claims without numeric content, and the excluded set is enriched for non-canonical phrasings where a mismatch is a priori *more* likely — so the exclusion was not conservative in a known direction. Worst case, had all twelve been unfaithful, the rate would have been $38/53 = 72\%$. §S6.6 lists the twelve exclusions verbatim and names the two regexes that produced them. That accounting is why the arm was rescored blind.

Blind rescore under a frozen rubric. The rubric (`p7_faithfulness_rubric.md`) — checkable assertions in any surface form, plus match rules including a base-grid convention for "grid plus additions" claims and a truncation convention for "with k removed" claims — was committed to git as `e181d2a` at 03:56:56 on 2026-08-01, **before** any rescoring run, so the freeze is checkable, and it pre-committed the reporting rule: below 90%, P-T4 reported not confirmed, full stop, no consolation framing. Scoring was blind — claim text and circles only, no predicted or rival values, no running tally, no threshold and no sight of the existing 38/41 result — with tallying afterward; all 60 labels are released verbatim in `p7_blind_labels.json`.

Result. Over all 60 trace_v2 rows: **54 MATCH, 2 MISMATCH, 4 UNSCOREABLE** — a corrected rate of $\mathbf{54/56 = 96.4\%}$ of scoreable claims, Wilson 95% CI [88%, 99%], passed, and on 56 scoreable claims rather than 41. The CI's lower bound sits just below the registered threshold, so "clears 90%" is a statement about the point estimate at this n. The two mismatches share a form: row 11 claims "alternating rows of 4 and 3 circles" over observed row sizes 4,3,3,3, row 33 claims "5 rows alternating between 5 and 4 circles" over observed 5,4,4,4,4 — both *mis-descriptions of alternation*, a repeating pattern named and built for the first period only. The original scorer's three mismatches were three different claim forms, not one; the filler-adds-rows mechanism the original scorer described was real, and the rubric's base-grid convention handles it, so all three score MATCH under hand adjudication.

Scope, three ways. The blind pass covers *invalid* rows too: the original 93% conditioned on completions that produced a correct packing — the wrong conditioning, since a method line attached to an overlapping layout is where description and artifact are likeliest to diverge — while the blind pass scored all 60 rows including the 7 invalid ones, 4 of the 60 unscoreable. Second, the check verifies the description against the *artifact*, not the process: the non-claim guard applies in full, and for a model already emitting grids, "5 × 5 grid" matching a 5×5 grid is closer to a self-consistency check than to the causal question the chain-of-thought literature asks (§6). Third, the audit is confounded with the elicitation arm, computed on the arm §S6.2 shows concentrates 87% of valid outputs onto one template, where correct description is close to free; faithfulness is informative precisely on off-template outputs, such as the pilot's triangular-hexagonal 6+5+4+3+2+1 sample, described exactly while summing to 1.75. The split by on- versus off-prediction is not run here (§8, stopping rule). What survives is narrow: in a domain where claims are checkable, the method lines this proposer emits are, at 54 of 56 scoreable claims, true of the object it produced — and both failures are a model describing a pattern it started and did not finish.

S6.6 The twelve exclusions of the original faithfulness scorer

§S6.5 states the consequence. The automated scorer's twelve exclusions were not claims lacking numeric content. Nine were regex coverage gaps: `DIMS_RE` matched only the literal `NxN` form, and `ROWSCOLS_RE` required rows before columns, so the following fell through despite carrying explicit checkable dimensions.

- "Regular 3 by 4 grid with one additional circle in row 4"
- "3 complete rows of 4 circles and 1 centered circle in the top row"
- "rectangular grid 4-4-4-1 with uniform radius 1/8"
- "Four-row grid with 4 circles in the first row and 3 in each of the remaining"
- "Rectangular grid arrangement with 5 columns and 4 rows, plus one additional circle on top edge"
- "Square grid 5 columns 4 rows with 1 additional circle at top"
- "Rectangular grid six columns by six rows radius one-twelfth with five circles removed"
- "Regular 5-column by 6-row grid plus one additional circle in the right margin"
- "Rectangular grid packing with five columns and six rows plus one additional circle"

Three further exclusions are closer to genuinely unscoreable: "Uniform horizontal strips with tight row spacing"; "Regular rectangular grid packing with corner circle"; "Equal-radius rectangular grid with 4 rows stacked vertically".

The nine gap cases are non-canonical phrasings, which is where a description is a priori *more* likely to diverge from the artifact. An exclusion rule that drops exactly those is not conservative in a known direction, and that is the reason §S6.5 reports the blind rescore over all 60 rows rather than the original 38 of 41.

S7 Arm B: the reference program, verbatim

The optimizer-alone baseline of §3.5 is one fixed program. It is reproduced here so that the claim that it is what a reader would write first can be checked rather than taken on assertion. Authorship is the manuscript's (Claude drafting under the human author, §8); version 1 was committed at `1932cc0` and this version, version 2, at `c486843`, both before the single scored run, and no result informed either. The runner substitutes `__N__` and `__SEED__` and hands the source to arm CL's scorer as if it were a model completion; the report asserts the template's SHA-256, `298ba71c9f20....` Version 1 differed in two places only: it bounded restarts by wall clock, importing `time` and `sys`, which the registered allowlist forbids, and was blocked at all 45 rows; version 2 fixes the count at 50.

```
# Arm B: the optimizer alone. A fixed reference program, written by the pipeline before any
# run and never tuned on a result, that does what a reader would write first: random-restart
# SLSQP over centre and radius variables, maximising the sum of radii under containment and
# pairwise non-overlap, for a fixed number of restarts. It is executed through arm CL's
# registered pipeline unmodified (python -I -S, the fixed driver, 120-second wall clock, one
# core, arm-F scoring, section 2.4's clearance rule), so its rows are scored exactly like the
# model-written programs it is the baseline for. Imports are the registered allowlist only
# (math, numpy, scipy); version 1 imported time and sys and was blocked (amendment 1).
#
# Usage inside the pipeline: arm_b_run.py substitutes N and SEED below and hands the source
# to arm_cl_analysis.score_row as if it were a model completion.
import numpy as np
from scipy.optimize import minimize

N = __N__
SEED = __SEED__
RESTARTS = 50   # amendment 1: fixed count; 80 restarts fit in 95 s at N = 31 on one core

rng = np.random.default_rng(SEED)
iu = np.triu_indices(N, 1)


def unpack(v):
    return v[:N], v[N:2 * N], v[2 * N:]


def objective(v):
    return -np.sum(v[2 * N:])


def objective_grad(v):
    g = np.zeros_like(v)
    g[2 * N:] = -1.0
    return g


def constraints(v):
    x, y, r = unpack(v)
    dx = x[iu[0]] - x[iu[1]]
    dy = y[iu[0]] - y[iu[1]]
    rs = r[iu[0]] + r[iu[1]]
    return np.concatenate([x - r, 1 - x - r, y - r, 1 - y - r, dx * dx + dy * dy - rs * rs])


def constraints_jac(v):
    x, y, r = unpack(v)
    m = len(iu[0])
    J = np.zeros((4 * N + m, 3 * N))
    ar = np.arange(N)
    J[ar, ar] = 1.0;            J[ar, 2 * N + ar] = -1.0
    J[N + ar, ar] = -1.0;       J[N + ar, 2 * N + ar] = -1.0
    J[2 * N + ar, N + ar] = 1.0;  J[2 * N + ar, 2 * N + ar] = -1.0
    J[3 * N + ar, N + ar] = -1.0; J[3 * N + ar, 2 * N + ar] = -1.0
    i, j = iu
    rows = 4 * N + np.arange(m)
    dx = x[i] - x[j]
    dy = y[i] - y[j]
    rs = r[i] + r[j]
    J[rows, i] = 2 * dx;        J[rows, j] = -2 * dx
    J[rows, N + i] = 2 * dy;    J[rows, N + j] = -2 * dy
    J[rows, 2 * N + i] = -2 * rs; J[rows, 2 * N + j] = -2 * rs
    return J


bounds = [(0.0, 1.0)] * (2 * N) + [(0.0, 0.5)] * N
cons = [{"type": "ineq", "fun": constraints, "jac": constraints_jac}]


def repaired(v):
    """Shrink every radius by the largest constraint violation so the packing is strictly
    feasible; SLSQP satisfies constraints only to its own tolerance."""
    x, y, r = unpack(v.copy())
    viol = max(0.0, -np.min(constraints(v)))
    r = np.maximum(r - viol - 1e-12, 0.0)
    return x, y, r


best_sum, best = -1.0, None
for _ in range(RESTARTS):
    v0 = np.concatenate([rng.uniform(0.1, 0.9, N), rng.uniform(0.1, 0.9, N), rng.uniform(0.02, 0.08, N)])
    res = minimize(objective, v0, jac=objective_grad, method="SLSQP", bounds=bounds,
```

```
                constraints=cons, options={"maxiter": 400, "ftol": 1e-12})
    x, y, r = repaired(res.x)
    if np.min(constraints(np.concatenate([x, y, r]))) < 0 or np.any(r <= 0):
        continue
    s = float(np.sum(r))
    if s > best_sum:
        best_sum, best = s, (x, y, r)

if best is None:
    print("[]")
else:
    x, y, r = best
    print("[" + ", ".join(f"[{a:.10f}, {b:.10f}, {c:.10f}]" for a, b, c in zip(x, y, r)) + "]")
```

S8 Transfer boundaries in full

§5 states each transfer arm's registered prediction, verdict and headline count in one page; this section is that text in full, with its tables.

The weak tier's template is a closed form, and §S9 reports the prediction in full: modal at all seven tested N, at four of five held-out N, under a perturbed container and under paraphrase. A characterization that sharp invites the inference that it describes the proposer inside a loop. Four registered arms each changed one thing about the setting — a parent, a generation, the serving path, the reasoning budget — and each carried registered predictions for the inference; two more changed the vendor. The persistence prediction failed in each of the first three: arm MU's falsifier fired, arm L failed two of its four predictions (a third was registered open), and arm P's predictions were unscoreable because validity collapsed, which arm P-D traces to the budget. This section reports them as results, not as caveats, because the same inference is routinely drawn from zero-shot characterizations of LLM proposers.

S8.1 Arm MU: the anchor dissolves after one conditioning step

Arm MU issues one-parent mutation calls: the bare prompt plus an existing packing and the instruction to modify it so the sum of radii increases. Three parents per cell ($N = 13, 21, 31$; $n = 15$ per condition-cell, 135 invocations): *A_anchor*, the model's own unconditioned anchor $T(k^*, N)$; *B_rival*, the provably better in-family rival $V(k^*{-}1, m)$; and *C_offfamily*, a diagonal-chain packing scoring ≈ 0.59–0.62, far below either. It was registered together with arm CH before sampling, with numeric decision thresholds, a power analysis (0.83 at the registered effect size), tie-against conventions, and a falsifier (F-MU1) whose consequence — the anchoring claim is scoped to *unconditioned* calls only — was written down before the first invocation.

F-MU1 triggered; the anchoring result dissolves under conditioning. Under B_rival the pooled anchor-rate is 0 of 26 valid (0%, CI $[0.00, 0.13]$) against a registered DISSOLVES threshold of $\leq 20\%$; keep-or-improve is 26 of 26 against a threshold of $\geq 50\%$. Every valid B_rival sample keeps or refines the rival construction, and every one inherits the rival's grid order (k-inheritance 26/26, disconfirming P-MU3's $< 50\%$ prediction). The template does not reassert itself when a better in-family parent is present. A_anchor makes the same point from the other side: even with the model's *own* anchor as parent and an explicit increase-the-sum instruction, only 7 of 28 valid samples (25%, CI $[0.13, 0.43]$) stay on the anchor — P-MU1's $\geq 50\%$ prediction is not met. The one condition where the template does reassert itself is the one the registration predicted it would: given the *off-family* diagonal parent, 20 of 34 valid samples (58.8%, CI $[0.42, 0.74]$) abandon the parent entirely and emit a k^*-grid construction — P-MU4 holds per-cell at $N = 21$ (7/13) and $N = 31$ (12/14), though not at $N = 13$ (1/7). A good in-family parent is followed, a bad off-family parent is discarded for the template: the attractor is real but weak, capturing the trajectory only when the parent is worse than what the template supplies.

Parent	Valid	Anchor-rate (CI)	Registered threshold	Keep-or-improve	k-inher.	Reading
B_rival (in-family argmax)	26	0/26 (0%, $[0.00, 0.13]$)	DISSOLVES if $\leq 20\%$	26/26 (bar $\geq 50\%$)	26/26	**F-MU1 triggered** — anchoring does not transfer to one-parent calls
A_anchor (model's own anchor)	28	7/28 (25%, $[0.13, 0.43]$)	P-MU1 predicted $\geq 50\%$	—	—	not met — even the anchor-as-parent does not hold the anchor
C_offfamily (diagonal, ≈0.6)	34	20/34 abandon parent for k^*-grid (58.8%, $[0.42, 0.74]$)	P-MU4: template reasserts	—	—	holds at $N = 21$, 31; not at $N = 13$ (1/7)

No conditioned output clears the family argmax (post hoc, disclosed). Arm MU's rows were scored against the *anchor*, so whether any conditioned output clears what the family can express was never asked of them at registration. It is asked here. Recomputing the family argmax from the closed form — 1.776142375 at $N = 13$, 2.258883476 at $N = 21$, 2.748528137 at $N = 31$ — arm MU's 135 invocations give 88 valid outputs at the primary tolerance and 63 at 10^{-9}. Eighteen of the 88 sit above the family argmax by more than the 5×10^{-8} granularity at which their own sums are recorded, by at most 1.5×10^{-6}, and *every one of the eighteen fails the* 10^{-9} *test*: they are packings whose overlap is small enough to pass the loose gate, and the excess is the overlap. Twelve further rows exceed the argmax only within that recording granularity, which is to say they are the argmax, written down. Arm CH's 45 invocations add 14 valid outputs at both tolerances, none above. The verdict rests on the 10^{-9} leg, where none clears, and it is corroborated at structure level: each of the eighteen carries an overlap the tighter gate rejects, and the excess is bounded above by the overlap. This re-reading adds no invocations to the study total; the recomputation is `arm_mu_ceiling.py`.

S8.2 Arm L: iterating the loop moves the anchor; nothing clears the family argmax

Every arm to this point samples once. The regime that discovery loops actually run in iterates: a proposal is scored, kept or discarded, and fed back as a parent. Arm L runs the minimal generational loop and reports the four registered predictions whether or not they held. The design is registered before sampling, with both prompt templates hashed and the stepper and scorer committed alongside. Generation 0 is the bare template A.1, byte-identical to arms F, M and CN; generations 1–5 are arm MU's registered mutation template verbatim, instantiated with the parent packing and its score. Two discriminating cells, $N = 13$ and $N = 31$, are crossed with two archive regimes — GREEDY, which retains only the single best valid proposal, and DIVERSE, which retains up to five distinct-valued ones — giving four lineages of six rounds at population 5: 120 invocations, of which 2 were runtime-rejected and counted. The parent for population slot i is archive[i mod |archive|], so the loop carries no sampling randomness of its own. All four lineages are evaluable under the registered floor. This is a minimal analogue and the registration says so: population 5, six rounds, one-parent mutation, no islands, no crossover, no program representation, and no evaluator signal beyond the parent's own score. Nothing here transfers to deployed systems, and no claim about any named system follows.

L2 confirmed; L1 and L3 not met. L2 (dissolution survives iteration) holds in both cells: the on-prediction rate falls from 4 of 9 valid generation-0 proposals to 5 of 20 conditioned ones at $N = 13$, and from 7 of 8 to 6 of 29 at $N = 31$. The falsifier F-L1 did not fire. L1 (the registered prediction is modal at generation 0 in every lineage) is **not met**, on one lineage: the 13-DIVERSE seed round returned four valid proposals split two on the prediction and two elsewhere, and the registered tie rule counts the tie against it. The other three lineages pass. The failure is a power artifact of the registration — we set four separate per-lineage bars over roughly four valid proposals each rather than one pooled bar per cell — and it is reported as scored, not rescued. L3 (loop diversity is scaffold-produced, so the greedy archive should return proposals toward the family) is **not met, with the direction reversed**: the conditioned on-prediction rate is 3 of 22 under GREEDY against 8 of 27 under DIVERSE. The comparison is confounded — greedy and diverse lineages differ in parent identity as well as archive width, and 13-GREEDY produced only 4 valid proposals across five conditioned rounds — so the reversal is reported as a disconfirmation of the registered prediction and not as evidence for its converse.

L4: no lineage clears the family argmax. L4 was registered open, with no direction predicted: report best-of-run against the family argmax. **No lineage clears it.** At $N = 13$ both lineages top out exactly at the family argmax 1.7761424; at $N = 31$ GREEDY reaches 2.6104167 and DIVERSE lands exactly on the family argmax 2.7485281. Across 49 valid conditioned outputs, zero exceed the best value the recipe family can express. Iteration moves the proposer off the anchor without moving it past the family *in these arms*. One correction is disclosed: the registered configuration stored the family argmax to seven decimals, and the clearance test compares at 10^{-9}; a proposal landing exactly on the rival at $N = 31$ therefore tested as *above* the truncated literal by 3.7×10^{-8}. The scorer now recomputes the argmax in closed form, agreeing with the stored value to 1.3×10^{-15}; the change is post-sampling and affects only the 10^{-9} clearance test (corrections ledger item 34). Scale is the honest limit: five conditioned generations, population 5, 49 valid conditioned outputs, one tier. The arm shows that this loop, at this size, on this channel, does not clear the family; it cannot show that a larger one would not. Raw rows in `arm_l_collect.jsonl`; frozen report `arm_l_report.json`, scored once on the complete ledger. The study corpus stood at 1068 invocations at that point.

S8.3 Arm P: the registered prompt on a bare, pinned serving path

Every direct-emission arm above ran through an agent runtime whose decoding parameters are not exposed and whose conditioning context the prompt digests lock only in part (§3.1). An external referee named that the paper's largest instrument hole. Arm P answers it the only way it can be answered: the registered prompts, byte-identical — the builder asserts every hash on file, and gives $N = 43$ its first — sent through a serving path that exposes and pins temperature (OpenRouter chat-completions, the vendor's own `claude-haiku-4.5` alias, temperature 1.0, no system prompt), at $n = 15$ per cell across the seven square cells, the five held-out cells and the three code cells: 225 invocations, 219 returning content and 6 refused by the serving path and recorded as such. It was registered with competing predictions on mode identity (P-P1, P-P2), on the code-channel clearance (P-P3, P-P4) and a falsifier on the discriminating-cell rate (F-P1), and it is a new arm on a new instrument, reported beside arms F, CN and CC and merged into none of them.

Validity collapsed. At the seven square cells, 0 of 105 outputs are valid at either tolerance. Every failure is an overlap: the responses are well-formed lists of the requested count with no prose and no fences, emitted with no reasoning tokens, and their median maximum pairwise overlap at $N = 13$ is 0.10, fifty times the value window. At the held-out cells, 4 of 75 are valid, all four at $N = 50$ and all four summing to 2.5, a full unit below the family argmax there (3.5295867); none clears. The direct-emission predictions are therefore UNSCOREABLE at every cell under the arm's own floor of five valid outputs, and F-P1's denominator is empty — the registration did not anticipate a collapse, and we report that rather than read a zero either way. On the code cells, arm CC's prompt byte-identical and executed and scored exactly as arm CC's (10-second timeout), the picture is the paper's: 12 valid programs of 45 (4, 6 and 2 by cell, only $N = 21$ above the floor) and none clears the family under §2.4's rule; P-P3 holds.

Two post-hoc diagnostics, disclosed as such. Temperature is not the cause: the $N = 13$ prompt at temperature 0.0, 0.5 and 1.0 gives 0 of 6 valid at each, all overlaps. The instrument is: the same prompt with the registered dispatch wrapper, through the agent runtime's haiku alias that arms F, CN and CC used, six subagents in parallel on the same day, gives 6 of 6 valid at both tolerances and lands the registered anchor $T(4, 13) = 1.625$ in 4 of 6, with one 3×3 grid carrying corner fillers at 1.6144 and one centre-plus-border construction at 1.7500, above the anchor and below the family argmax.

Arm P-D: which component of the instrument. Registered after an outside review asked for a diagnostic beyond temperature (`arm_pd_preregistration.txt`, commit `5554cc0`, pushed before the first call): the same $N = 13$ prompt, same alias, same pinned path and decoding, under two one-line manipulations, $n = 15$ each. D1 turns the vendor's extended thinking on (the routing layer's default effort; median 3160 reasoning tokens per row) with no system prompt; D2 leaves thinking off and adds one system line, "You are a careful assistant. Think before you answer." Competing predictions: P-PD1, D1 valid in at least 8 of 15 and D2 in at most 3 (the reasoning budget is the cause); P-PD2, the mirror image (the context is); P-PD3, anything else.

The arm closed at 27 of 30 rows (14 D1, 13 D2) by a registered amendment after the serving path refused the last three for credit; they could not have changed the verdict, since D2 can reach at most 2 valid and D1 already holds 12. Result: **D1 is valid in 12 of 14** (11 at 10^{-9}; the two failures overlap), **D2 in 0 of 13**, all overlaps, exactly arm P's collapse. P-PD1 holds. And the anchor comes back with the budget: the modal valid D1 output is $T(4, 13) = 1.625$ (5 of 12, the 4×4 grid truncated), $k = 4$ is the modal structure (6 of 12), and one output is the family argmax 1.7761424 itself, a construction the runtime rows never reach at this cell (0 of 18 valid; both runtime rival hits are at $N = 21$). S-PD1 is met.

What the arm establishes. The anchoring result is a property of the instrument that produced it, and arm P-D names the component: the reasoning budget. The same prompt, sent bare to the same vendor's alias with temperature pinned, does not yield valid packings from this model class at all; give the pinned call the vendor's extended thinking and it yields valid packings at the runtime's rate with the template modal, while a system line that asks for care changes nothing. The runtime's undisclosed conditioning context — the fragment §3.1 disclosed the digests do not lock — is not what carries the template; the thinking budget is. That is why Table 1 carries an instrument column and never pools across it, and it is a reproducible instrument, since the budget is a request parameter on the pinned path. Ledgers and scripts are listed in Appendix E.

S8.4 Across vendors: a boundary that runs both ways

Two registered arms send the same byte-identical prompts across vendors; the verdict table, arm GM3 per cell and arm V in full are Supplement S2. **Arm GM3** runs the unconditioned-call protocol against a second vendor's open-weight family (gemma-4-26b-a4b-it) through its first-party API, seven N cells at twenty samples each, predictions and falsifier fixed before sampling. The pooled prediction held — 46 of 80 valid packings (57.5%) land within 2×10^{-3} of the value predicted by §2.3's rule, against a registered 30% bar — and the pass rides on the non-discriminating cells: on the three discriminating cells the rate is 11 of 43 (25.6%), below the bar; the cell-level prediction did not hold (modal in 2 of 5 scoreable cells, short of the registered 5) and the falsifier (failure in ≥ 4) did not trigger either, so the registration left a dead zone and the outcome landed in it. The three misses are the finding §4.2 uses: the family's modal output at every discriminating cell is the rival argmax itself, 27 of 43 discriminating-cell validities carrying its exact two-radius decomposition. **Arm V** varies vendor and serving class at once, thirteen free-tier aliases at five cells and five samples each (877 attempt rows over 325 registered slots). Ten of thirteen aliases finished below the registered validity floor and claim nothing in either direction. Of the three that cleared it, two transfer at point estimate — 8/12 and 11/18 on-prediction, Wilson intervals spanning the 50% bar — and gemma-4-31b does not, at 0/15. Restricted to the cells that can discriminate anchoring from family search the decomposition gives 3 of 4 and 5 of 9, and neither separates from the template-shape null of §S9.2 (exact upper tails 0.111 and 0.145); zero rival-argmax emissions in 16 discriminating-cell validities.

So the anchor is neither a single-vendor artifact nor universal: it transfers at the pooled level to two further vendors, one weak-tier family takes the rival instead, and a sibling of that family at the same floor anchors on nothing. With arms MU, L and P this closes the boundary from every side the study reached: the template is a property of one tier, one channel, one instrument, and one conditioning regime, and it is inside those four conditions that the closed form names the modal output in advance.

S9 The mode prediction and the arms that bound it

The closed form that names the weak tier's modal output in advance, reported in full; the body keeps only what Table 1 and §4 need from it.

S9.1 Trap zones and the bound table

Substituting $k^*(N)$ into the branch condition: TRAP $N \in [k^2 - k + 1, k^2 - 1]$, CONVERGE $N \in [k^2, k^2 + k]$. Over $N = 10 \ldots 60$: $[13, 15]$, $[21, 24]$, $[31, 35]$, $[43, 48]$ and $[57, 63]$ clipped to $[57, 60]$ by the sweep bound.

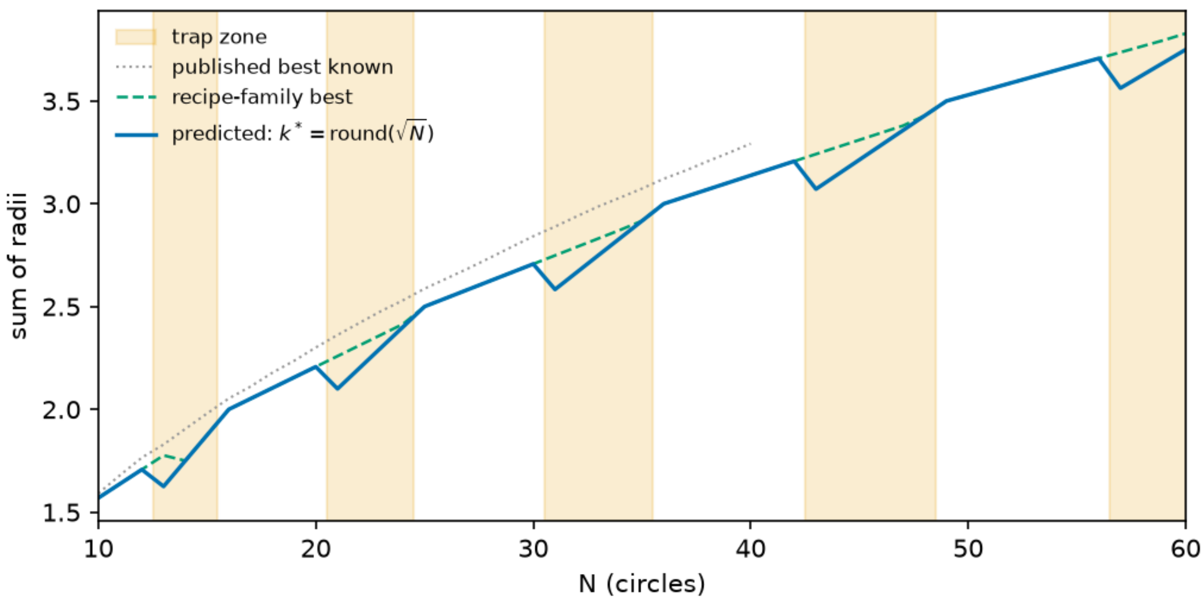

Figure S3: *Predicted versus optimal value,* $N = 10\ldots60$. (i) the rule's prediction; (ii) the best value in the recipe family; (iii, dotted) published best-known values (Friedman, n.d.), terminating at $N = 40$ because the source page stops there. Shaded bands are the trap zones, the last being $[57, 63]$ clipped by `sweep(10,60)`. Worst-in-zone penalties 8.51%, 7.03%, 6.01%, 5.25%, 4.66%; these are family-internal arithmetic and exact at every N, while the published-bound comparison (curve iii) is checkable only up to $N = 40$, where the bound table stops. The gap closes to zero at $N = 35$ and $N = 48$ but **not** at $N = 24$, where 0.59% remains — curve (ii) sits strictly above (i) across $[21, 24]$.

Inside a zone the rule generally loses value against the best construction available *within the recipe family*. Worst-in-zone penalties fall as N grows: 8.51% at $N = 13$, 7.03% at 21, 6.01% at 31, 5.25% at 43, 4.66% at 57. The penalty reaches zero at $N = 35$, 48, 14 and 15, but **not** at $N = 24$, where truncation gives $T(5, 24) = 2.4000000$ against $V(4, 8) = 2.4142136$, a residual 0.59%. Branch label and penalty are therefore not coextensive: 4 of the 22 trap-branch N in range cost nothing, so "trap" names the branch, not a guaranteed loss. Where the penalty is zero, truncation is separable from convergence only by structure; §S9.3 uses $N = 35$ as that control.

The linear-programming verification summarized in §2.2 originally covered $k = 2\ldots7$ and stopped short of two registered cells, arm M's $T(8, 57)$ and arm CN's predictions at $k = 8$ and 9; the sweep was extended rather than the claim narrowed (`lp_extend_k89.py`), and every registered closed-form value is inside the verified range.

Our bound table (`n_sweep_forecast.json`, transcribed from Friedman (n.d.)) covers $N = 10\ldots40$; through the round before this one it stopped at $N = 30$, although the source page has listed $N = 31$ since 2011 and, as fetched on 2026-09-02, runs to $N = 40$ (corrections ledger, row 35). Its $N = 26$ entry, 2.63598, matches ShinkaEvolve's figure truncated. That entry has since been re-credited on the source page to a July 2026 submission, so we withdraw the reading that LLM-driven systems hold the record here and claim only what the table supports. What survives is that the recipe family is never competitive with that record: the family's best value sits 0.02–0.16 below it across $N = 10\ldots40$ (the rule's own value 0.02–0.31), and 0.0946 at $N = 26$ ($2.63598 - 2.5414214 = 0.0945586$). Above $N = 40$ there is no bound table, so both the deficit claim and the LP gate's "never exceeds a published bound" abort are unchecked there, covering two of five trap zones and one of our seven square cells ($N = 43$).

S9.2 The mode prediction: seven cells, all modal

An exact point prediction invites a fair objection: a per-sample hit rate well short of 100% is not "predicting the exact output". The number should be read against the sampling entropy of the distribution, computed in `p11_mode_baseline.json` (bare arm, 2×10^{-3} buckets). At **all seven tested N the predicted value is the empirical modal output** — four discriminating cells plus three where the prediction and the family argmax coincide, so the anchoring-specific evidence sits in the four — and the on-prediction rate equals the modal frequency to within one sample. The two columns are counted on different partitions: the modal-frequency column groups at four decimals, the on-prediction column uses the registered 2×10^{-3} window. They come apart at exactly one cell, $N = 31$, where the window admits one packing the four-decimal bucket separates out, which is why that row reads 12/17 against 13/17:

N	predicted	empirical mode	modal freq	on-prediction / valid	k^*-structure / valid†
13	1.6250000	1.6250	10/18	10/18 (56%)	17/18
17	2.0517767	2.0518	3/4	3/4 (75%)	3/4
21	2.1000000	2.1000	12/15	12/15 (80%)	13/15
31	2.5833333	2.5833	12/17	13/17 (76%)	17/17
35	2.9166667	2.9167	3/4	3/4 (75%)	3/4
37	3.0345178	3.0345	3/4	3/4 (75%)	4/4
43	3.0714286	3.0714	6/7	6/7 (86%)	7/7

† Post-hoc structural check, not preregistered, run after the square arm closed (`diagnostics_kmatch.py`): the grid order backed out of each sample's dominant radius ($k = \mathrm{round}(1/2r)$) equals $k^*(N)$. All 50 of 50 on-prediction samples are k^*-structured — the value match is not an accounting coincidence — and 64 of 69 valid samples overall (93%) sit on the k^* grid, so most value misses are k^*-grid variants (perturbed radii or filler counts), not different constructions.

No predictor of a single value can exceed the modal frequency, and this one attains it at 7 of 7 cells: the residual is dispersion, not misprediction.

The modal-frequency argument holds only if the mode and the on-prediction count are read on the same partition, so the two partitions in play have to be separated. On-prediction is the registered 2×10^{-3} value window. The modal-bucket column is grouped at four decimals, a reporting choice, and the two differ at exactly one cell: at $N = 31$ on-prediction is 13 of 17 against a 4-dp modal bucket of 12 of 17. Read on the registered window — the partition the modal-frequency claim is about — the mode is 13 and the predictor attains it. The 4-dp column is kept because it is what the per-cell tables display, not because any verdict rests on it. Three cells ($N = 17$, 35, 37), all non-discriminating, carry four valid samples each, where a 3/4 mode is thin evidence on its own; the well-populated cells of the held-out probe (§S9.4) are the stronger test of the same rule. As a floor, a naive "the model emits a round number" baseline hits 2 of the same 69 valid samples (3%). A stronger baseline, disclosed post hoc: a proposer choosing uniformly among the family's plausible template shapes — one branch value per order $k \in \{k^*{-}1, k^*, k^*{+}1\}$ — lands on-prediction 1/3 of the time; the observed discriminating-cell rates sit above that null with exact binomial upper tails 0.043, 2.9×10^{-4}, 3.4×10^{-4} and 6.9×10^{-3} at $N = 13$, 21, 31 and 43 (`diagnostics_template_null.py`). All four survive Holm correction at $m = 4$, whose thresholds are 0.0125, 0.0167, 0.025 and 0.05 — including $N = 13$, the weakest, which an earlier draft of this section reported as uncorrected. The 7-of-7 mode identity does not depend on this null. Two scope conditions: this is the bare weak-tier arm — pooled over the four discriminating square cells (full ledger), the two higher tiers and the rectangle arm (arm G, Supplement S5), the on-prediction rate is 46% (47/102, pooling the counts rather than the rates: $41 + 1 + 0 + 5$ on-prediction over $57 + 30 + 4 + 11$ valid; the three non-discriminating square cells, 9/12, are excluded because prediction equals the family argmax there), because §3.3's higher tiers are almost never on-prediction — and "modal" is a property of the observed sampling regime, not of a pinned decoding configuration (§7).

Which on-prediction number is which. The rate is reported on several corpora and cell sets, each for a reason; every one is tabulated here against its own denominator, so that no later mention has to be read twice.

figure	counts	corpus	cells
56–86%	on-prediction	bare weak-tier arm	each of the seven square cells
18/23	on-prediction	original corpus	the four discriminating square cells
41/57	on-prediction	full shipped ledger	the same four
2/23, 2/57	rival emissions	those two corpora	the same four
9/12	on-prediction	full shipped ledger	the three non-discriminating square cells
12/16	on-prediction	original 45-row arm	the three matched cells (13, 21, 31)
35/50	on-prediction	full matched-cell ledger	the same three
47/102	on-prediction	full ledger plus both higher tiers and the rectangle arm (Supplement S5)	the four discriminating square cells, plus those tiers and that container

The headline figure for the anchoring claim is **41/57**: the full shipped ledger at the four cells where prediction and family argmax differ. Everything else is a subset, a superset, or a different question.

One label belongs on that figure at the point of use rather than only in §3.1. Of its four cells, two — $N = 21$ and $N = 43$ — carry no individually registered point prediction; their values follow deterministically from a formula fixed before sampling, but a reader auditing what was locked should know which is which. Dropping $N = 43$, whose prompt hash is also unregistered (§8), gives 35/50. Restricting to the two cells carrying both a registered hash and a registered prediction, $N = 13$ and $N = 31$, gives **23/35**. The rate degrades from 72% to 70% to 66% across those three readings, so nothing turns on the choice; we quote 41/57 and state 23/35 beside it.

S9.3 Square container, original corpus and full ledger

The square arm sampled weak-tier proposers at $N = 13$, 17, 21, 31, 35, 37 and 43. The rule was fitted on $N = 23/26/27$ only, so every cell is out of sample. Four cells are *discriminating*: rule and family optimum disagree, so a proposer that searched the family would return a higher number. Across those cells ($N = 13$, 21, 31, 43), **18 of 23 valid invocations landed on the predicted construction** and the rival-argmax value was reached **2 times in 23** — both at $N = 21$, both the 4×4 grid plus five fillers at 2.2588835.

Those figures cover the **original 45-invocation bare arm only** (ids ≤ 5 at $N = 13/31$, ≤ 10 at $N = 21$, all rows at $N = 17/35/37/43$), the corpus that existed when §3 was written. The trace study (Supplement S6) later added 40 bare rows at $N = 13/21/31$ under the same arm label; over the full shipped ledger the same four cells give **41/57 on-prediction and 2/57 rival**. Both are reported. §3 and the trace study are computed on nested subsets of one arm, not on independent samples, and Supplement S5 carries the split. The 40 added rows are the bare arm of the trace study (Supplement S6): 60 bare samples were collected there, of which 20 are the pre-existing arm-F rows — $N = 13$ ids 1–5, $N = 21$ ids 1–10, $N = 31$ ids 1–5 — and 40 are new, so the 57 valid rows behind 41/57 are the original arm plus those 40 and nothing else.

That the anchor first breaks at $N = 21$ is suggestive — it is the bottom of a zone carrying a 7.03% penalty, so the anchor looks weak where obeying it costs — but that is a hypothesis, not a finding. **P4** registered $N = 37$ as converging on 3.0345178, a prime predicted clean, contradicting the prior work's guess; it confirmed. **P5** registered $N = 35$ at 2.9166667, the top-of-zone control; three of four valid samples landed there, separable from convergence only because the structure classifier reads radii rather than totals; the fourth used a 7×7 lattice truncated to 35 (sum 2.5), outside the rule. At $n = 4$ this is "consistent with" rather than confirmed, and deserves running to $n = 20$.

Bookkeeping. Five invocations were rejected by the runtime's 20-subagent concurrency cap *before reaching a model*. We excluded them under a rule that was not preregistered, so the arm is reported both ways — $35/45 = 77.8\%$, or $35/50 = 70.0\%$ had the rejections been scored as model failures — and the five records appear in the ledger in no form (§8). The arm also contains three parse failures, and one $N = 37$ proposer derived $r = (\sqrt{2} - 1)/12$ correctly in prose, then transcribed 0.03571429.

S9.4 Arm CN: the rule predicts the mode where there is nothing to recall

The rule was fitted on $N = 23/26/27$, with $N = 26$ the canonical published cell, so a rival reading survives every square-arm result above: the proposer recalls packings it saw rather than constructing a round($\sqrt{N}$) grid. Arm CN samples the bare template A.1 at five N absent from the sum-of-radii scoreboard cited in §6 and from every prior arm — $N = 50$, 58, 62, 65 and 75, mirroring the square arm's branch mix (three truncation cells, two $m = 1$ extend cells; the $m \geq 4$ branch is already disconfirmed by arm M). Absence from the pretraining corpus cannot be established and is not claimed. Two predictions were registered against each other before sampling (pushed first; motivation disclosed as post-review): P-CN1, construction — the registered value is modal at ≥ 4 of 5 cells including ≥ 2 of the 3 discriminating cells; P-CN2, recall — modal at ≤ 2 of 5. Fewer than five valid outputs is UNDERPOWERED; ties count against. $n = 15$ weak-tier invocations per cell, scored once by the arm-F pipeline unchanged.

N	k^*, prediction	disc.	valid [Wilson 95%]	on-pred.	mode (count, margin)	verdict	k^*-struct.
50	7, $V(7, 1) = 3.5295867$	no	11/15 [48–89%]	4	3.5122554 (6, +2)	**MISS**	10/11
58	8, $T(8, 58) = 3.6250000$	yes	14/15 [70–99%]	12	3.6250000 (12, +11)	**HIT**	14/14
62	8, $T(8, 62) = 3.8750000$	yes	13/15 [62–96%]	13	3.8750000 (13, +13)	**HIT**	13/13
65	8, $V(8, 1) = 4.0258883$	no	13/15 [62–96%]	9	4.0258871 (9, +7)	**HIT**	12/13
75	9, $T(9, 75) = 4.1666667$	yes	12/15 [55–93%]	8	4.1666667 (8, +6)	**HIT**	8/12

P-CN1 holds. The registered value is modal at 4 of 5 cells, at all three discriminating cells, with margins of 6 to 13 samples; every cell clears the evaluability floor; the family rival is emitted 0 times in 39 discriminating-cell validities and no valid output exceeds the family argmax (0/63). The registered structural secondary holds at 5 of 5 cells: the dominant radius backs out $k = k^*$ in 57 of 63 valid outputs. The one miss, $N = 50$, is the arm-CH pattern: 10 of 11 valid outputs are the registered 7×7 grid plus one filler, 4 at the registered interstitial radius $(\sqrt{2} - 1)/14$ and 6 with it in a container corner at $r \approx 0.0123$ — the cell's mode, 0.017 below the prediction. The template is selected, the filler mis-executed, and the miss is reported as one. At $N = 75$ the four off-mode outputs are alternative constructions, none of them the rival $V(8, 11)$. Validity is 63/75 (84%) with no collapse at large N; the invalid rows are nine overlaps, two count errors and one output of fraction literals the registered parser does not evaluate (the CH "radical literal" mode).

The $N = 65$ reconstruction. Raw rows are in `arm_cn_collect.jsonl` (75/75 launched, none lost). Eleven $N = 65$ rows were transcribed by reconstructing the standard 8×8 grid string from the emitted filler triple, flagged per row (`reconstructed: true`) and disclosed in `arm_cn_results.txt`; the reconstruction is character-identical for these standard grids. One of the eleven is invalid, so discarding them removes ten of the cell's thirteen valid rows and leaves three — below the five-row floor, which makes the cell unscoreable and P-CN1 3 hits of 4 evaluable cells. The three discriminating cells are untouched either way. The arm brings the study corpus to 678 invocations.

What this does and does not show. The anchor survived the move to N with nothing to recall at four of five cells and structurally at all five, so the closed form describes what the proposer builds, not what it has seen, at the branches where it was already supported. This does not establish that the fitted cells are absent from training text and does not extend to the $m \geq 4$ branch; the perturbed-container and direct-recall probes are now arms CP and RP (§S9.5), and the iteration question is now arm L (§S8.2).

S9.5 Arms CP, RP and PP: the anchor without its words, recall asked directly, and paraphrase

Arms CP and RP were drafted at review stage (draft files committed 2026-08-22, disclosed as such) and registered together; arm PP followed. Each registration carries its prompt hashes and analysis script, pushed before the first invocation, and each arm was sampled once and scored once under its own stopping rule, on the same weak-tier subagent channel as arms F, M and CN.

Arm CP destroys every lexical handle while preserving the geometry: the container becomes $[3, 5] \times [3, 5]$, so the model never sees "unit square", "[0,1]", or any published magnitude, while the similarity map $x \mapsto 2x + 3$ leaves the optimum, the family and the branch rule invariant (predictions are exactly $2\,V$ or $2\,T$; circles are mapped back and scored under the arm-F conventions unchanged). Five cells, all discriminating, 15 each. P-CP1 (modal valid output equals the mapped prediction at ≥ 4 of 5 evaluable cells) **confirmed**: all five cells clear the five-valid floor and the mapped prediction is modal at four — $N = 13$ (valid 14/15, on-prediction 9), 21 (13/15, 7), 31 (9/15, 8) and 58 (12/15, 11). At $N = 75$ (13/15 valid) the mode ties, five outputs at the mapped $T(9, 75) = 8.3333333$ against five at the mapped $T(15, 75) = 5.0$, a deeper truncation of the *same* family; the registered tie rule scores the cell a miss. Pooled validity is 81%, far above the 40% collapse branch of the lexical-keying alternative P-CP2, and the structural secondary S-CP2 holds at 5 of 5 cells ($k = k^*$ majority). The strong form S-CP1 (rival-argmax emissions ≤ 1) is **not met**: 4 of 61 discriminating-cell validities emit the mapped rival, two at $N = 13$ and two at $N = 21$ — 6.6%, against 9 of 261 = 3.4% over every other weak-tier arm in the study pool, defined next. The falsifier F-CP1 (lexical keying) did not fire. One handle survives by construction: N itself, the index any memorized table would be keyed by.

The pooled rival rate, both ways (post hoc, disclosed). Over the weak-tier discriminating-cell validities of the original prompt lineage and the cross-vendor zero-shot arm, the provably better rival is emitted **13 times in 322**, 4.0%: arm F 2/57, trace (arm T, Supplement S6) 1/53, PP 6/75, CP 4/61, CN 0/39, arm V (Supplement S2) 0/16, rectangle (arm G, Supplement S5) 0/11, arm M 0/10. That rule excludes arm GM3, which is weak-tier and direct-emission too and emits the rival 27 times in its 43 discriminating-cell validities (§5); folding it in gives **40 of 365, 11.0%** — nearly triple. We report both figures and no longer describe the narrower pool as covering every arm in the study. The inclusion rule is the instrument: the pooled arms share the arm-F prompt lineage, and GM3 runs the GM-chain instrument at twenty samples per cell against this pool's five to twenty.

Three further disclosures. The pool is an unregistered aggregate — each constituent verdict is registered, the aggregate is not — over arms with different bars, containers and cell sets, so it is quoted descriptively and never tested. The two arms carrying most of the narrower count, CP and PP, are exactly the two whose registered strong-form secondaries are not met; a pool dropping them reads 3 in 186, which we do not headline because the arms it drops are the ones that failed. The code-channel arms CC and CC2 sit outside both pools; folding them in gives 15 of 400, 3.75%, slightly *below* the rate we report.

Arm RP asks the recall question outright: "What is the best-known maximum sum of radii for $\{n\}$ non-overlapping circles packed inside the unit square?", answer as a number or UNKNOWN, at three cells with a published value in the paper's own citations ($N = 13$, 26, 30; the draft named $N = 32$, replaced before registration by $N = 30$ because no cited value exists at 32, documented in the registration) and three held-out cells ($N = 50$, 62, 75). **Every one of the 87 scored responses answers UNKNOWN** — 0 of 44 scoreboard-cell responses recall a published value (bar for the recall branch P-RP2: ≥ 10), zero state a family value, zero state anything else; three invocations were runtime-rejected and counted. P-RP1 **confirmed**, the family-rate asymmetry trivially zero, and the registered falsifier F-RP1 (P-RP2 satisfied) did not fire. A second-round review noted the design carried no positive control, so uniform UNKNOWN could reflect an abstention prior induced by offering the option; a registered amendment closed this (pushed before its sampling): at $N = 1$, where the optimum 0.5 is elementary, the same prompt draws the correct number in **14 of 15** responses (one UNKNOWN) — C-RP1. The UNKNOWNs therefore reflect the absence of a statable value under this prompt, not the format. The registered caveat stands: a model that fails to state a value may still have been shaped by it, which bounds the simplest mechanism only.

Arm PP holds the container and varies the words: three semantic-preserving paraphrases of template A.1 (clause reorder with synonyms; a declarative numbered-list rewrite; compact mathematical phrasing), fixed and hashed, registered before sampling; the paraphrases were authored by the pipeline that had seen all prior results, disclosed there. Two discriminating cells, $N = 13$ and 31, 15 invocations each. P-PP1 **confirmed at 6 of 6**: the registered prediction is the modal valid output at every paraphrase-cell (valid 11–14 of 15 per cell, pooled validity 83%, on-prediction 4–12), so the anchor is not keyed to the hash-locked prompt string — §8's (model, prompt string) scoping weakens to the task level at these cells. The strong form again does not carry: 6 of 75 valid outputs emit the rival against the registered S-PP1 bar of ≤ 2; $k = k^*$ majority holds at 5 of 6 cells. A formatspread-style sweep (Sclar et al., 2024) remains out of scope.

References

Erich Friedman. Packing center: circles in squares, maximizing the sum of radii. `https://erich-friedman.github.io/packing/cirRsqu/`, n.d. Best-known values transcribed into `n_sweep_forecast.py`; accessed

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